

Functional Sheet Loopy

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1 Introduction

Loopy is a full working armband that works like a nowadays smartwatch with some peculiarities. The armband can register different data, like:

- **Movement Activity**
- **Hearth Rate**
- **Sweating**
- **Blood Oxigenation**
- **Body Temperature**
- **Sleep Levels**
- **Stress**
- **Blood Glucose**

As sad before we can measure with our device all the data a normal smartwatch can measure, but with some improvements.

Why choose Loopy instead of a normal smartwatch? The Loopy armband is more usable and safe, can retrieve best data and define a comfort during the use also in long period of usage.

Loopy application works like a *remote* for the device and permits to visualize datas in real time and to manage the device status.

Loopy is also very modern with the use of AI during the calculus of some data and the usage of the AI Loopy Chatbot integrated into the application.

2 Functional Requirements

2.1 FR001 - Create an Account

Loopy application defines *authentication* methods. Before to start with the usage you need to create an account with some of your data:

- **Username**
- **Email**
- **Password**
- **Height**
- **Age**
- **Weight**

You can setup all of these into the register page.

2.2 FR002 - Usage of the Loopy Device

The armband is already setuped with your account and, intead of all other devices like this, when you boot it up and put it on you arm it starts to get data.

2.3 FR003 - Application Usage

The application shows all the datas the device get from your day. It's divided into 5 pages:

- **Home Page:** shows your account and some *graphs* of **Sleep**, **Stress** and **Activity** of your days. It also shows here you last detected hearth rate and daily consumption.
- **Data Page:** more specific than Home Page it shows in numbers all of the data extracted from you daily activities during the use of Loopy Device in a more *numerical way* for more precize datas.
- **Chat Page:** meet Loopy! An AI Agent designed only on your data. You can ask Loopy some descriptions of your datas and what you can do to enhance your lifestyle. You can see a Loopy message also in Home Page that gives you a summaried checkup in base of your data.
- **Device Manager:** this page defines all the statuses of the different sensor of the device to find some errors during the Loopy Device usage.
- **Profile:** here you can check your profile status and the data you have saved during the registration.

2.4 FR004 - Delete an Account

On the settings page you can find a button for the account deletion if you want to reset you datas with the creation of another account or stop the Loopy Usage.

3 Non-Functional Requirements

3.1 NFR001 - Loopy Device Performance

Loopy device can provide you the last information in milliseconds using the application, you can check device statuses and data in real time without waiting.

3.2 NFR002 - Loopy AI Performance

Loopy can provide you informations with messages with a maximum wait of 6-7 seconds. It provides accurate informations with the help of Google Gemini Flash LLMs.

3.3 NFR003 - Security

All of your personal informations are crypted during in the full usage of loopy with no informations leaks or information losses.

3.4 NFR004 - Compatibility

Loopy can be used by anyone, the data will be stored also without the usage of the mobile application. The mobile application instead is usable only from Android phones (no iPhone and no Desktop).